Vision World Models for Competitive Snake Playing

Team 5

March 2026

It is sometimes claimed that we are close to having AI with "superintelligence".

> Time is short, and we must accelerate our actions to match accelerating AI progress. Possibly by 2026 or 2027 (and almost certainly no later than 2030), the capabilities of AI systems will be best thought of as akin to an entirely new state populated by highly intelligent people appearing on the global stage—a "country of geniuses in a datacenter"—with the profound economic, societal, and security implications that would bring. There are potentially greater economic, scientific, and humanitarian opportunities than for any previous technology in human history—but also serious risks to be managed.

Arguably, we barely have any AI with "human-level intelligence". For example:

- A 10 year old can often learn to fill up a dishwasher within **minutes**.
- A 17 year old can often learn to drive a car within **tens of hours**.
- Yet we (still) do not have level 5 "fully-autonomous" self-driving cars with **millions of hours** of training data.

How might machines learn like animals?

(L)LMs are popular now, but limited by construction. Consider the standard transformer (Vaswani et al., 2017):

- It does a fixed amount of computation per token at inference.
- It does not model the consequences of its actions at inference.
- If it is autoregressive with a non-zero probability $\varepsilon \in \mathbb{R}_{[0,1]}$ of error (by generating a "bad" token) at each generation step, and errors are independent across steps, then the probability of generating an error-free token sequence decreases exponentially with sequence length $n \in \mathbb{N}$ because:

$$(1 - \varepsilon)^n \approx \exp(-\varepsilon n)$$

- The real world often has nothing to do with language (Einstein, 1931); humans use discrete, low-bandwidth, and clear symbols for communication, but the real world is often continuous, high-bandwidth, and noisy (LeCun, 2025).

World models have been proposed as an approach to human-level AI.

Multiplayer Snake has been put forward as a research request by OpenAI.

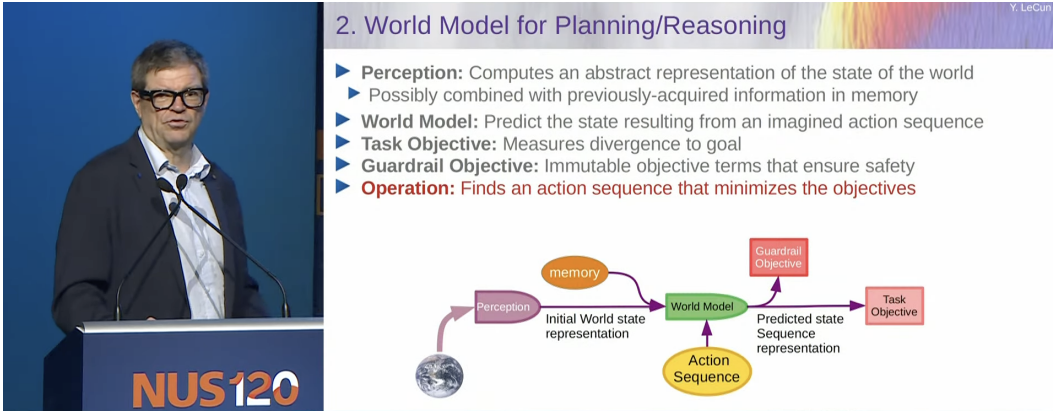
Requests for Research

⭐⭐ **Slitherin'.** Implement and solve a multiplayer clone of the classic Snake game (see slither.io for inspiration) as a Gym environment.

- Environment: have a reasonably large field with multiple snakes; snakes grow when eating randomly-appearing fruit; a snake dies when colliding with another snake, itself, or the wall; and the game ends when all snakes die. Start with two snakes, and scale from there.
- Agent: solve the environment using self-play with an RL algorithm of your choice. You'll need to experiment with various approaches to overcome self-play instability (which resembles the instability people see with GANs). For example, try training your current policy against a distribution of past policies. Which approach works best?
- Inspect the learned behavior: does the agent learn to competently pursue food and avoid other snakes? Does the agent learn to attack, trap, or gang up against the competing snakes? Tweet us videos of the learned policies!

As our research contribution, we investigate vision world models in the simple setting of competitive Snake.

What we shall discuss has been discussed in many works, including talks by LeCun.



We identify 2 published works which, as far as we can tell, sit closest to our work in task and method:

- **Multiplayer Snake ... using Self-Play** (Francis, 2019) has our task but uses PPO.
- **TD-MPC** (Hansen et al., 2022) has our method but uses continuous-control DMControl tasks.

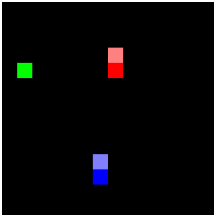
Data

We can simulate the game to collect (image, action, next image, cost) tuples in a dataset:

```
initialise snakes
initialise food
initialise time
initialise dataset

while a snake is alive:
 translate each snake
 grow any snake on food

 mark as dead each snake whose head is on a wall or body
 for each cell with multiple heads:
  mark as dead each snake but random one from longest ones in cell
 remove each snake marked as dead

 respawn food if eaten
 add to dataset (image, action, next image, cost)
 increment time
```

Data

cell	colour
background	black
body of playable snake	blue
head of playable snake	ceil(mean(blue, white))
body of non-playable snake	red
head of non-playable snake	ceil(mean(red, white))
food	green



We can colour and render the grid at arbitrary size using rasterisation.

Data

We can numerically encode the preference of "avoiding death beats pursuing growth".

For example, the cost at time $t \in \mathbb{N}$ can be a numerically encoded preference

$$c_t = \left(1 + \left\lceil \frac{1}{1-\gamma} \right\rceil\right) d_t - g_t$$

where $\gamma \in \mathbb{R}_{[0,1]}$ is a discount factor, and d_t is 1 if death occurred at time $t+1$ (and 0 otherwise), and g_t is 1 if growth occurred at time $t+1$ (and 0 otherwise).[1]

[1] The coefficient derives from the series $\sum_{t=0}^{\infty} \gamma^t = \frac{1}{1-\gamma}$.

Data (for rigour)

Let $\mathrm{L} \in \mathbb{N}_{\geq 1}$ be a grid extent, and $\mathrm{U} \in \mathbb{N}_{\geq 1}$ be an upsampling factor, and $\mathrm{S} \in \mathbb{N}_{\geq 1}$ be a number of snakes.

$$\mathcal{G} = [1..\mathrm{L}]^2 \tag{grid}$$

$$\mathcal{A} = \{(0,0), (0,1), (0,-1), (-1,0), (1,0)\} \tag{action space}$$

$$\mathcal{M} = \mathcal{A} \setminus \{(0,0)\} \tag{movement space}$$

$$\mathcal{S} = \bigcup_{n \geq 2} \{\boldsymbol{s} \in (\mathbb{Z} \times \mathbb{Z})^n \mid \forall i \in [1..n-1](s_{i+1} - s_i \in \mathcal{M})\} \tag{snake space}$$

$$\mathcal{F} = \mathcal{G} \cup \{\perp\} \tag{food space}$$

$$\delta : \mathcal{S} \times \mathbb{N} \to \mathcal{A} \tag{action oracle}$$

$$\mathcal{O}(\boldsymbol{S}) = \{s_i \mid \boldsymbol{s} \in \boldsymbol{S} \wedge i \in [1..|\boldsymbol{s}|]\} \tag{snake occupancy}$$

$$\rho : \mathcal{G}^{2\mathrm{S}} \to (\mathcal{G} \times \mathcal{G})^{\mathrm{S}}$$

$$\rho(\boldsymbol{x}) = ((x_{2i-1}, x_{2i}))_{i=1}^{\mathrm{S}} \tag{pairer}$$

Data (for rigour)

Let $\mathcal{X} = \mathbb{N}_{\leq 255}^{\mathrm{UL} \times \mathrm{UL} \times 3}$ be an image space, and $\mu_{\mathrm{B,w}} = (128, 128, 255)$ be light blue, and $\mathrm{B} = (0, 0, 255)$ be blue, and $\mu_{\mathrm{R,w}} = (255, 128, 128)$ be light red, and $\mathrm{H} = (255, 0, 0)$ be red, and $\mathrm{G} = (0, 255, 0)$ be green, and $\mathrm{b} = (0, 0, 0)$ be black.

$$\mathcal{C} : \mathcal{G} \times \mathcal{S}^* \times \mathcal{F} \to \mathbb{N}_{\leq 255}^3,\ \mathcal{C}((i,j), \boldsymbol{S}, f) = \begin{cases} \mu_{\mathrm{B,w}}, & \text{if } |\boldsymbol{S}| \geq 1 \wedge (i,j) = s_{1,1} \\ \mathrm{B}, & \text{if } |\boldsymbol{S}| \geq 1 \wedge \exists k \in [2..|\boldsymbol{s}_1|]\big((i,j) = s_{1,k}\big) \\ \mu_{\mathrm{R,w}}, & \text{if } \exists r \in [2..|\boldsymbol{S}|]\big((i,j) = s_{r,1}\big) \\ \mathrm{H}, & \text{if } \exists r \in [2..|\boldsymbol{S}|], \exists k \in [2..|\boldsymbol{s}_r|]\big((i,j) = s_{r,k}\big) \\ \mathrm{G}, & \text{if } f \neq \perp \wedge (i,j) = f \\ \mathrm{b}, & \text{otherwise} \end{cases} \tag{painter}$$

$$\mathcal{D} : \mathcal{S}^* \times \mathcal{F} \to \mathcal{X},\ \mathcal{D}(\boldsymbol{S}, f) = \left(\mathcal{C}\left(\left(\left\lceil \frac{i}{\mathrm{U}} \right\rceil, \left\lceil \frac{j}{\mathrm{U}} \right\rceil\right), \boldsymbol{S}, f\right)\right)_{i,j \in [1..\mathrm{UL}]} \tag{renderer}$$

Data (for rigour)

$\boldsymbol{S} \sim \rho \circ \mathit{lu}(\{\boldsymbol{s} \in \mathcal{G}^{2S} \mid \forall i \in [1..S](\boldsymbol{s}_{2i} - \boldsymbol{s}_{2i-1} \in \mathcal{M}) \wedge |\{\boldsymbol{s}_i\}_{i=1}^{2S}| = 2S\})$
$E \leftarrow \mathcal{G} \setminus \mathcal{O}(\boldsymbol{S})$
if $E = \varnothing$ **then**
  $f \leftarrow \perp$
else
  $f \leftarrow \mathit{lu}(E)$
end if
$t \leftarrow 0$
while $|\boldsymbol{S}| > 0$ **do**
  $i_f \leftarrow 0$
  for all $\boldsymbol{s} \in \boldsymbol{S}$ **do** ▷ Translate snakes
    $d \leftarrow \delta(\boldsymbol{s}, t)$
    if $d = (0, 0)$ **then**
      $h \leftarrow 2\boldsymbol{s}_1 - \boldsymbol{s}_2$
    else
      $h \leftarrow \boldsymbol{s}_1 + d$
    end if
    $\boldsymbol{s} \leftarrow h \parallel (\boldsymbol{s}_i)_{i=1}^{|\boldsymbol{s}|-1}$
  end for
  for all $\boldsymbol{s} \in \boldsymbol{S}$ **do** ▷ Grow snakes
    if $\boldsymbol{s}_1 = f$ **then**
      $\boldsymbol{s} \leftarrow \boldsymbol{s} \parallel (2\boldsymbol{s}_{|\boldsymbol{s}|} - \boldsymbol{s}_{|\boldsymbol{s}|-1})$
      $i_f \leftarrow 1$
    end if
  end for
  $D \leftarrow \varnothing$
  $B \leftarrow \{\boldsymbol{s}_i \mid \boldsymbol{s} \in \boldsymbol{S} \wedge i \in [2..|\boldsymbol{s}|]\}$
  for all $\boldsymbol{s} \in \boldsymbol{S}$ **do**
    if $\boldsymbol{s}_1 \notin \mathcal{G} \vee \boldsymbol{s}_1 \in B$ **then**
      $D \leftarrow D \cup \{\boldsymbol{s}\}$
    end if
  end for
  $H \leftarrow \{\boldsymbol{g} \in \mathcal{G} \mid \exists \boldsymbol{c} \subseteq \boldsymbol{S}(|\boldsymbol{c}| \geq 2 \wedge \forall \boldsymbol{s} \in \boldsymbol{c}(\boldsymbol{s}_1 = \boldsymbol{g}))\}$
  for all $h \in H$ **do**
    $C \leftarrow \{\boldsymbol{s} \in \boldsymbol{S} \mid \boldsymbol{s}_1 = h\}$
    $w \leftarrow \mathit{lu}(\arg\max_{\boldsymbol{s} \in C} |\boldsymbol{s}|)$
    $D \leftarrow D \cup (C \setminus \{w\})$
  end for
  for all $d \in D$ **do** ▷ Kill snakes
    $\boldsymbol{S} \leftarrow \boldsymbol{S} \setminus \{d\}$
  end for
  if $i_f = 1$ **then** ▷ Respawn food
    $E \leftarrow \mathcal{G} \setminus \mathcal{O}(\boldsymbol{S})$
    if $E = \varnothing$ **then**
      $f \leftarrow \perp$
    else
      $f \leftarrow \mathit{lu}(E)$
    end if
  end if
  $t \leftarrow t + 1$
end while

Architecture

$$\boxed{\text{Encoder}} \longrightarrow \boxed{\text{Action-conditioned latent world model}} \longrightarrow \boxed{\text{Energy}}$$

The encoder and world model are learnable; the energy can be learnable or non-learnable:

- The encoder predicts a hidden representation from an image.
- The world model predicts the next hidden representation from a hidden representation, action, and latent representation.
- The energy predicts a scalar cost measuring how "good" a hidden representation is.

Like how young animals in an environment might first internally learn representations (by passive observation of environmental evolution) before dynamics (by active environmental interaction), learning could occur in stages:

1. The encoder could be trained alone.
2. The encoder could be trained together with the world model.
3. The encoder and world model could be trained together with the energy.

Architecture (for rigour)

Let $\mathrm{H} \in \mathbb{N}_{\geq 1}$ be a hidden space dimensionality, and $\mathrm{Z} \in \mathbb{N}_{\geq 1}$ be a latent space dimensionality, and $\mathcal{H} = \mathbb{R}^{\mathrm{H}}$ be a hidden space, and $\mathcal{Z} = \mathbb{R}^{\mathrm{Z}}$ be a latent space, and $\mathbb{P}_{\mathcal{Z}}$ be a probability measure on $\mathcal{Z}$.

$$\phi : \mathcal{X} \to \mathcal{H} \tag{encoder}$$

$$\mathscr{F} : \mathcal{H} \times \mathcal{A} \times \mathcal{Z} \to \mathcal{H} \tag{world model}$$

$$\mathscr{C}, \mathscr{V} : \mathcal{H} \to \mathbb{R}_{\geq 0} \tag{energy}$$

We can train $\mathscr{V}$ on $\sum_{i=t}^{\tau-1} \gamma^{i-t} c_i$ and $\mathscr{C}$ on c_t.

The solution can be given by a discounted sum planner like

$$\mathcal{U}\left(\arg\min_{a \in \mathcal{A}^T} \mathbb{E}_{z \sim \mathbb{P}_{\mathcal{Z}}^{\otimes T}}\left[\gamma^T \mathscr{V}\left(\left(\bigcirc_{i=1}^{T} \mathscr{F}(\cdot, a_i, z_i)\right)(\phi(x))\right) + \sum_{t=1}^{T} \gamma^{t-1} \mathscr{C}\left(\left(\bigcirc_{i=1}^{t} \mathscr{F}(\cdot, a_i, z_i)\right)(\phi(x))\right)\right]\right)$$

where $x \in \mathcal{X}$ is an image, and $\gamma \in \mathbb{R}_{[0,1]}$ is a discount factor, and $T \in \mathbb{N}_{\geq 1}$ is a planning horizon.

Project Timeline Plan

- Week 9: Data-related implementation.
- Week 10: Model-related implementation.
- Week 11: Model training and evaluation.
- Week 12: Report and slide deck preparation.

Project Task Distribution

- Celest: Training and testing of model(s).
- Ryan: Implementation of model(s).
- Yihe: Implementation of data functions.
- Gregory: Writing for report and slide deck, and so on.

Appendix

We use the following conventions.

Let $n \geq 1$, and X be a set, and $f_1, \ldots, f_n : X \to X$.

$$\bigcirc_{i=1}^{n} f_i = f_n \circ \cdots \circ f_1$$

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